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Harmonizing of Experiment and Computation Model for Estimating Cell Performance and Water distribution inside of Polymer Electrolyte Membrane

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In this study, we estimated the performance of PEMFC as changing operating temperature in different inlet humidity condition at cathode side but anode dry, and tried to match experimental results with 1-dimensional simulation to construct database for advanced computation model. We used Nafion®112 membrane and a self-manufactured PEMFC with 25cm^2 active area was used. The range of operating temperature was 40~70℃ and oxygen through bubbled humidity chamber was supplied 0~80% humidity condition as changing water temperature in humidity chamber. For figuring out governing equations express water contents in electrolyte membrane, the linear forward difference method was applied about time progress and quadratic central difference method was used about space progress. In low operating temperature condition, 40~60℃, increasing temperature rarely effected cell performance but we can see performance drop at 70℃. By modifying Henry's constant and/or diffusion coefficient, the modified one-dimensional model was accomplished for calculation.

Introduction

The proton exchange membrane fuel cell is one of the most interesting research topics these days, because PEMFCs are considered promising as a device for clean and efficient energy conversion with the highest power density at lowest operation temperature and polymer electrolyte fuel cells are being recognized as a promising candidate for mobile applications such as zero-emission electric vehicles (1-3). One for the major problems, however, is the requirement of maintaining the membrane electrolyte in humid conditions (the so-called water management), because the electric conductivity depends strongly on the water content in the membranes (4, 5).

Mass transfer in PEMFCs is one of the important concerns effecting on performance and it is influenced by water contents in an electrolyte membrane. Therefore, it is great issue to control produced water caused by electrochemical reaction and predict water contents in electrolyte membrane for high efficiency and durability in PEMFCs. In case of two and three dimensional simulation, it takes lots of time and cost for analyzing whole process and these computation models can't predict a change of water contents in electrolyte membrane. On the other hand, one dimensional model can efficiently estimate the cell performance in short time because it is able to predict the effect as changing of water contents in spite of it can't obtain the information about characteristics of flow channel, gas diffusion layer, catalyst layer. Okada et.al (6, 7) accomplished numerical analysis about effect on electrolyte membrane's Na pollution and water contents inside of electrolyte as applying 1-dimensional diffusion equation at electrolyte area. Also, 1-dimensional computation was achieved about flow channel, gas diffusion layer and electrolyte area by Springer (8) et.al. They assumed that the electrolyte membrane's inside was completely hydrated and fully humidified at anode and cathode sides. These results from ideal conditions, however, are different from real experimental results.

For estimating the performance of PEM fuel cell correctly, new 1-dimensional simulator development related with laboratory scale to include real losses is required. The objective of present study is to construct database for predicting the cell performance and percentage of water content in electrolyte membrane at laboratory scale condition. That is, this study is a part of whole processes to develop advanced PEMFC simulator for estimating the accurate cell performance. Therefore, the performance of PEMFC was estimated as changing operating temperature and relative humidity at cathode side, and modified 1-dimensional model was matched with experimental results as comparing calculated results from new developed model obtained by modifying Henrry's constant and diffusion coefficient with experimental ones.

Experiment & Models

Water transport in electrolyte membrane

Figure 1 represents water transport mechanism in PEMFC. Water is supplied to the fuel cell as humidified condition, and hydrogen is disassembled protons and electrons by oxidizing reaction with catalyst layer. After H+ ion is transported from anode to cathode side by electro-osmotic reaction, water is produced by-products reducing reaction with oxygen. Back diffusion of water generates from cathode to anode side by water concentration difference between electrodes, and water transport is also appeared between two electrodes because of convection according to pressure gradient.

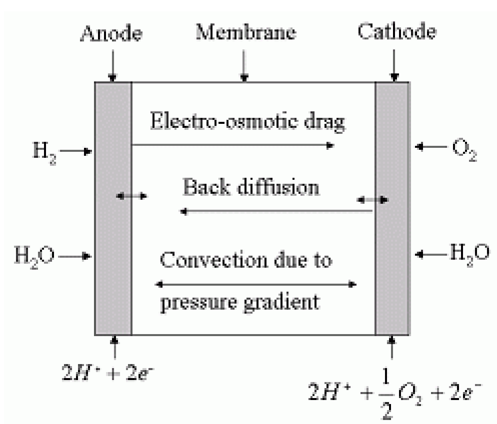
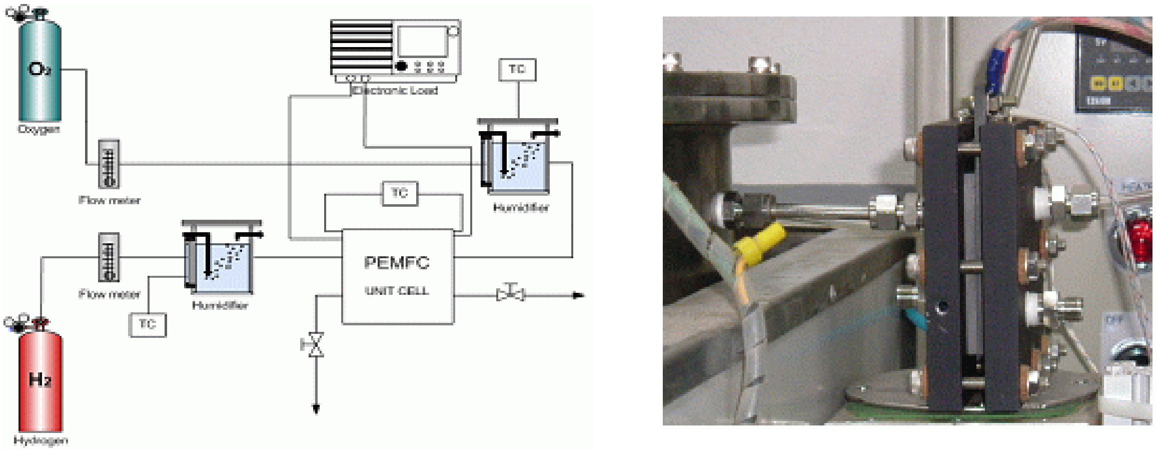


Figure 1 Schematic of the water transport process in PEMFC

Experimental setup and method



(a) (b)

Figure 2 (a) Schematic diagram of experimental setup and (b) PEM Unit cell on test system

Figure 2 shows schematic diagram of PEMFC's experiment system (a) and a unit cell used in this paper (b). We used Nafion®112 membrane and carbon papers (1.0mg/cm Pt loading, 20wt.% Pt/Vulcan XC-72, E-TEK). A self-manufactured PEMFC with active area of 25 ㎠ was used. Pure hydrogen and general (normal) oxygen were used as fuels, and electric load (KIKUSUI inc.) was used to get I-V curve.

We estimated the performance of a PEMFC's unit cell as changing operating temperature in different inlet humidity conditions at cathode side but anode dry, and tried to match experimental results with 1-dimensional simulation. The operating temperature in an unit cell was controlled in 40~70℃ through heater controller attached at end-plate, and hydrogen and oxygen were supplied through bubbled humidity chamber. At the same time, inside temperature of MEA was measured by thermocouple.

1-dimensional model

In this study, absorption (evaporation) term was added because water at cathode side is humidified, and it is expressed in Eq. (1)

$$\frac{\partial}{\partial t}\left(\frac{\lambda dx}{2}\right) = h_c(\lambda_c - \lambda) + D\frac{\partial \lambda}{\partial x} - \frac{EW}{\rho}\frac{\eta_{drg}\cdot I}{F} - \frac{EW}{\rho}\frac{K_p}{\mu}\lambda\frac{\partial p}{\partial x} \qquad at \quad x = 0 \qquad (1)$$

Also, Eq. (2) is applied because water is produced by oxygen reducing reaction at cathode side.

$$\frac{\partial}{\partial t}\left(\frac{\lambda dx}{2}\right) = h_c(\lambda_c - \lambda) - D\frac{\partial \lambda}{\partial x} - \frac{EW}{\rho}\frac{\eta_{drg}\cdot I}{F} + \frac{EW}{\rho}\frac{K_p}{\mu}\lambda\frac{\partial p}{\partial x} + \frac{EW}{\rho}\frac{I}{2F} \quad at \quad x = 0 \quad (2)$$

TABLE 1. Equations used in calculation of cell voltage

Equation	Expression	Reference
E	$1.229-0.85\times10^{-3}(T-298.15)+4.3085\times10^{-5}T(\ln(p^{*}_{H_2})+0.5\ln(p^{*}_{o_2}))$	10
V_{act}	$V_{act}=\xi_1+\xi_2 T+\xi_3 T[\ln(c_{o_2})]+\xi_4 T[\ln(I)]$, $\xi_1=-0.944$ $\xi_2=-0.00354$, $\xi_3=7.80\times10^{-5}$, $\xi_4=-0.000196$, $c_{o_2}=\dfrac{p_{o_2}}{5.08\times10^{6}\exp(-\frac{498}{T})}$	9
V_{con}	$V_{con}=I(c_1\dfrac{I}{2.2})^2$ $c_1=\begin{cases} if\ \dfrac{p_{o_2}}{0.1173}+p_{sat}\prec 2 \\ \quad (7.16\times10^{-4}T-0.622)(\dfrac{p_{o_2}}{0.1173}+p_{sat})- \\ \quad 1.45\times10^{-3}T+1.68 \qquad else \\ \quad (8.66\times10^{-5}T-0.068)(\dfrac{p_{o_2}}{0.1173}+p_{sat})- \\ \quad 1.6\times10^{-4}T+0.54 \end{cases}$	10
V_{ohm}	$V_{ohm}=I\dfrac{L}{\sigma_m}$, $\sigma_m=(\dfrac{\lambda_m}{29.81+\lambda_m}-0.06)^{1.5}(\dfrac{\lambda_1^0}{1-5.5})\dfrac{1}{18\lambda_m}\alpha$ $\lambda_1^o=349.8\exp(-\dfrac{14000}{8.3143}(\dfrac{1}{T}-\dfrac{1}{298}))$ $\alpha=\dfrac{\lambda_m+1-\sqrt{(\lambda_m+1)^2-4\lambda_m(1-1/K_{AC})}}{2(1-1/K_{AC})}$ $K_{AC}=6.2\exp(\dfrac{52300}{8.3143}(\dfrac{1}{T}-\dfrac{1}{298}))$	11

Water diffusion coefficient and the percentage of water content is present following.

$$\begin{cases} 0.043+17.81a-39.85a^2 \\ +36.0a^3 \qquad for 0\prec a\leq 1 \\ 14.0+1.4(a-1) \qquad for 1\prec a\leq 3 \end{cases} \tag{3}$$

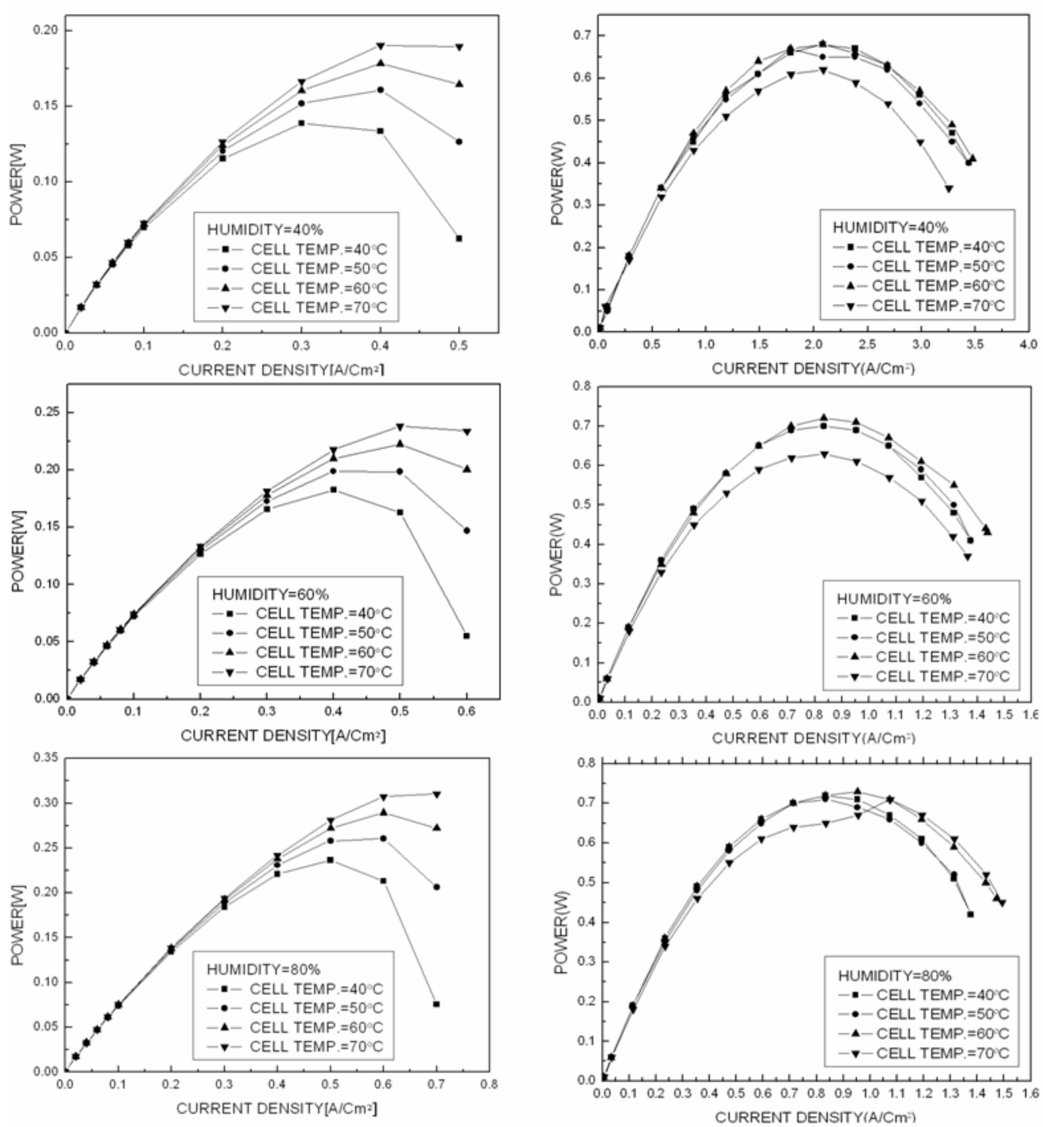
$$\begin{cases} 3.1\times10^{-3}\lambda(e^{0.282\lambda}-1)\bullet e^{(-2346/T)} \\ \qquad for 0\prec\lambda\leq 3 \\ 4.17\times10^{-4}\lambda(1+161e^{-\lambda})\bullet e^{(-2346/T)} \\ \qquad otherwise \end{cases} \tag{4}$$

Cell voltage is defined as:

$$V_{cell} = E - V_{act} - V_{con} - V_{ohm} \tag{5}$$

The empirical formulas by Mann (9), Serra (10), and Chan (11) are used for calculating activation voltage, concentration overvoltage and ohmic voltage of electrolyte membrane. Table 1 shows the empirical formulas used. For figuring out governing equations, represent water contents in electrolyte membrane, the linear forward difference method was applied for time progress and quadratic central difference method was used for space progress. It was assumed that pressure terms were linearly changed due to thin electrolyte membrane. The boundary condition was set up that produced water was exhausted when water contents in electrolyte membrane exceed permissible amount.

Results and Discussions



(a) (b)

Figure 3 (a) The I-P curve using 1-dimensional model and (b) The I-P curve from experimental results: as changing operating temperature (40℃, 50℃, 60℃, 70℃) at constant humid condition (40%, 60%, 80%)

Table 2. Inside temperature of MEA in OCV condition.

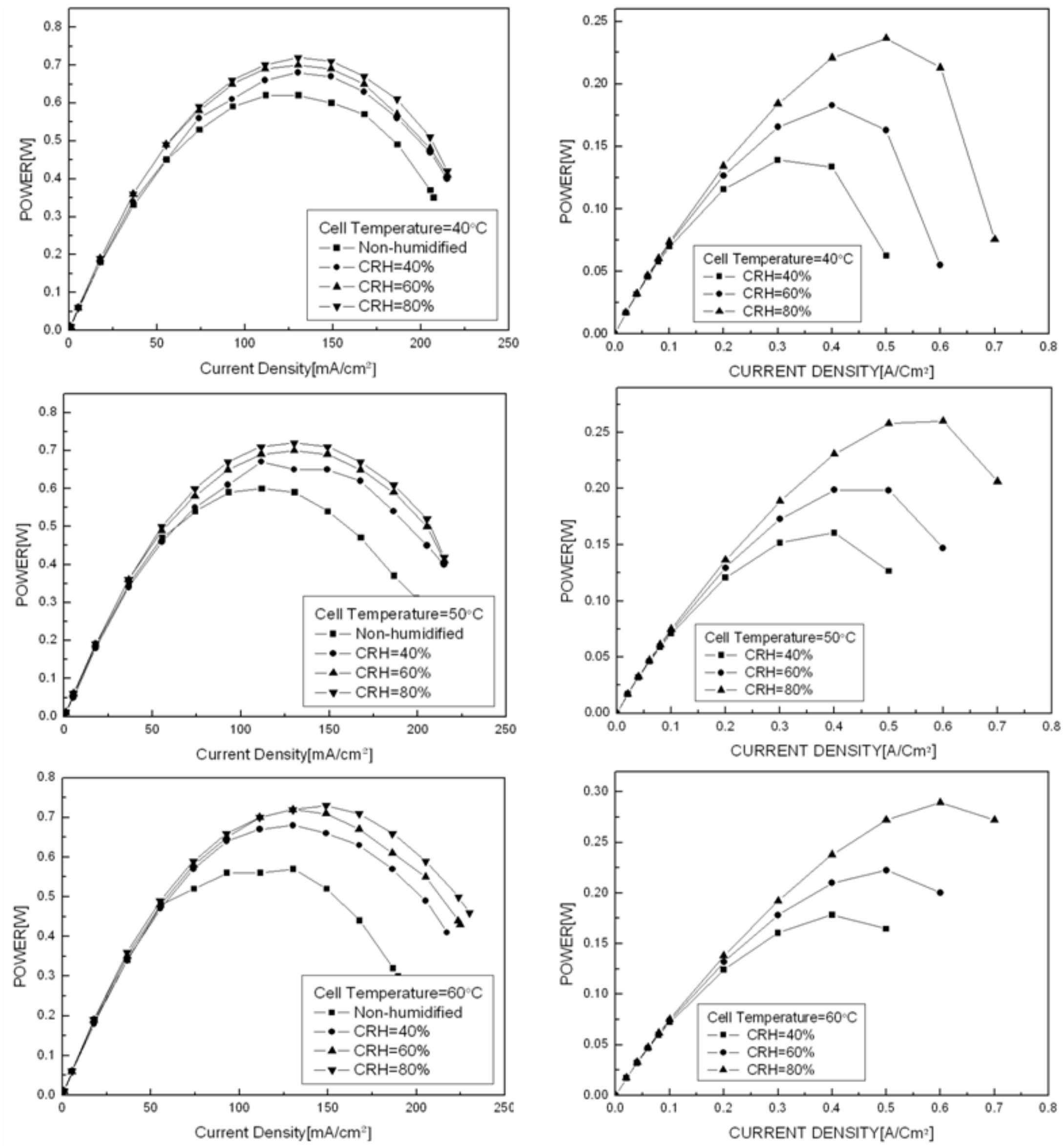
Operating temperature(℃)	Inside temperature of MEA(℃)
40	90
50	106
60	120
70	131

Figure 3 shows the performance of PEMFC as changing operating temperature in fixed relative humidity at cathode inlet side. 1-dimensional computation results are expressed at Figure 3 (a). Cell performance is improved as increasing operating temperature at different humidity conditions because the increasing operating temperature makes the cell resistant decreased due to high ion conductivity and reaction rate, and mass transfer rate. Figure 3 (b) is experimental result under same conditions. In low operating temperature, 40~60℃, increasing temperature rarely effected cell performance but we can see performance drop at 70℃.

To investigate the reason, the thermocouple was inserted between carbon papers during MEA manufacturing process and inside temperature of MEA in OCV condition was measured at various operating temperature. Table 2 represents the measured internal temperature of MEA. As table 2 shows, when operating temperature is 70℃, inside temperature of MEA increases to 130℃ because of electrochemical reaction between hydrogen and oxygen. Therefore, we can guess this voltage drop is ascribed to damaged membrane because the used Nafion® 112 membrane is generally damaged over 120℃.

Figure 4 presents I-P curve as changing relative humidity(0%~80%) of cathode inlet side at different operating temperature (40℃, 50℃, 60℃). 1-dimensional analysis results are expressed at Figure 4 (a). The performance of PEMFC is increased according to higher inlet humidity condition at cathode side because high relative humidity of the oxygen caused water concentration to increase in electrolyte membrane and high water concentration makes ion conductivity of electrolyte membrane increasing. Figure 4 (b) shows experimental results at same operating conditions. Humidified oxygen compared with non-humidified state makes cell performance higher and the cell performance is increased according to rise relative humidity at cathode inlet side.

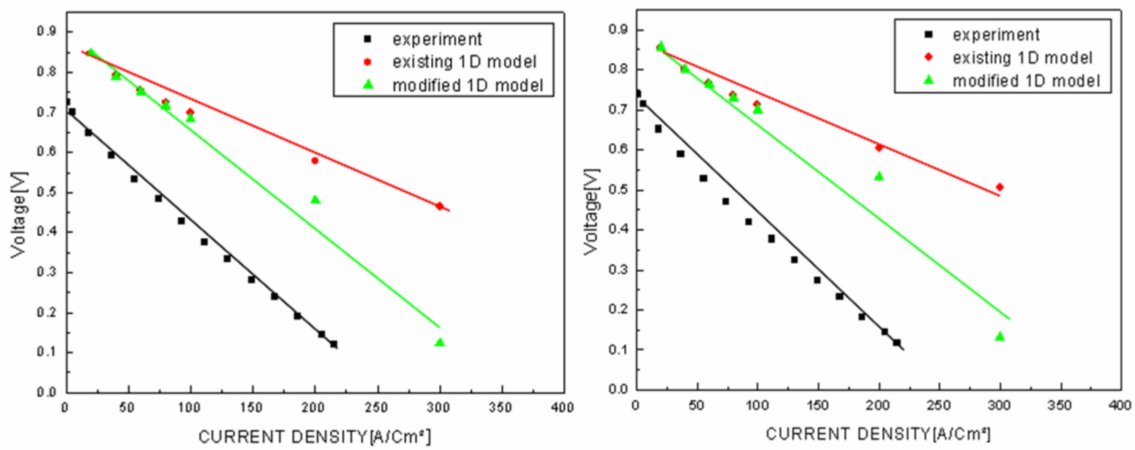
In low operating temperature region, maximum output power is higher 16% than non-humidified condition while output power is higher 30% in high operating temperature region. The more operating temperature is increased, dry rate of electrolyte membrane is higher and it causes ion conductivity drop. At this moment, supplying humidified oxygen makes the water content of electrolyte membrane increase and on this account ion conductivity, reaction rate, and mass transfer rate are increase, therefore output power is improved.



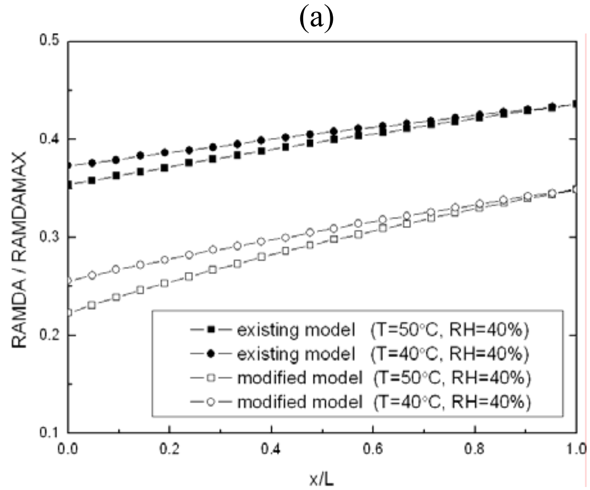
(a) (b)

Figure 4 (a) The I-P curve using 1-dimensional model as increasing relative humidity (40%, 60%, 80%) at different operating temperature (40℃, 50℃, 60℃) and (b) The I-P curve from experimental results as changing humid condition (0%, 40%, 60%, 80%) at fixed operating temperature (40℃, 50℃, 60℃)

Figure 5 (a) compares with experiment and one-dimensional calculation results at operating temperature 40℃, 50□ and relative humidity 40% respectively. Previous one-dimensional model (12) was accomplished in ideal condition but in this work, absorption (evaporation) term based on Henrry's constant and water production term caused by oxidizing reaction was added. Then, computation results from advanced model were acquired by modifying Henrry's constant and diffusion coefficient. As Figure 5 shows, the slope of Modified one-dimensional model shows good agreement with that of experiment in low humidified region but the absolute value is different because of experimental errors such as measurement loss of MEA, contact pressure of unit cell, and humidity rate of fuel. Figure 5 (b) represents changing of the percentage of water content at internal electrolyte membrane using existing and modified



(a)



(b)

Figure 5 (a) The I-V curve comparisons with existing and modified model at operating temperature: 40□, 50□ and relative humidity: 40% and (b) the percentage of water content inside of electrolyte membrane using 1-dimensional simulation.

model when current density is 0.3A/㎠ at same operating condition as (a). The experiment and 1-dimensional model was matched as changing the percentage of water content by modify Henry's constant and diffusion coefficient. Additional study, however, is needed to predict performance of PEMFC in wide range of operating temperature and humidity condition and reduce the experimental errors.

Conclusion

In this paper, estimating the performance of PEMFC was accomplished as changing operating temperature and relative humidity at cathode side and we matched new model, modified Henry's constant and diffusion coefficient from existing 1-dimensional model, and experimental results to construct database for developed 1-dimensional simulator. According to increase relative humidity, the performance of PEMFC is higher due to improved ion conductivity of electrolyte membrane. The cell performance was decreased at high operating temperature region because of damaged electrolyte membrane by high internal temperature. The slope compared to experiment and modified 1-dimensional model is almost same at low relative humidity region. However, it needs to study applicable 1-dimensional model at broad operating region for developing advanced computation model to predict the cell performance correctly.

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